

ML model training and testing results

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Author	Sean Winklaar

Environment & Training Configuration

All models evaluated in this research were configured using the ultralytics framework. To ensure a fair baseline comparison, the models were initialized with pre-trained weights and trained using identical hyperparameters targeting the tiled dataset structure.

Before running the training script, the dataset was systematically partitioned to ensure unbiased evaluation and explicitly monitor for overfitting. We split the data in 3 folders, splitting the image patches and their corresponding YOLO text annotations into distinct **Training**, **Validation**, and **Testing** subsets using a **70%, 20%, 10%** distribution ratio.

- Training folder: 294 Images.
- Validation folder: 69 Images.
- Testing: 30 Images (Unseen data).

Parameter Definitions

- **epochs=30**: The training duration was restricted to 30 epochs based on the hyperparameter tuning phase, where model accuracy stabilized and reached optimal convergence without overfitting.
- **imgsz=512**: Defines the resolution matching our image tiling pre-processing step. This forces the model to look at sharp, high-contrast local features rather than downscaling full drone canvases.

Summary of Data Strategy

We first tested our starting model (**YOLOv8n**) on regular, large images (non tiled), it completely failed to make good predictions. This is a common problem in drone and aerial imagery: when big images are shrunk down to fit into the model, small target objects become blurry and hard to see.

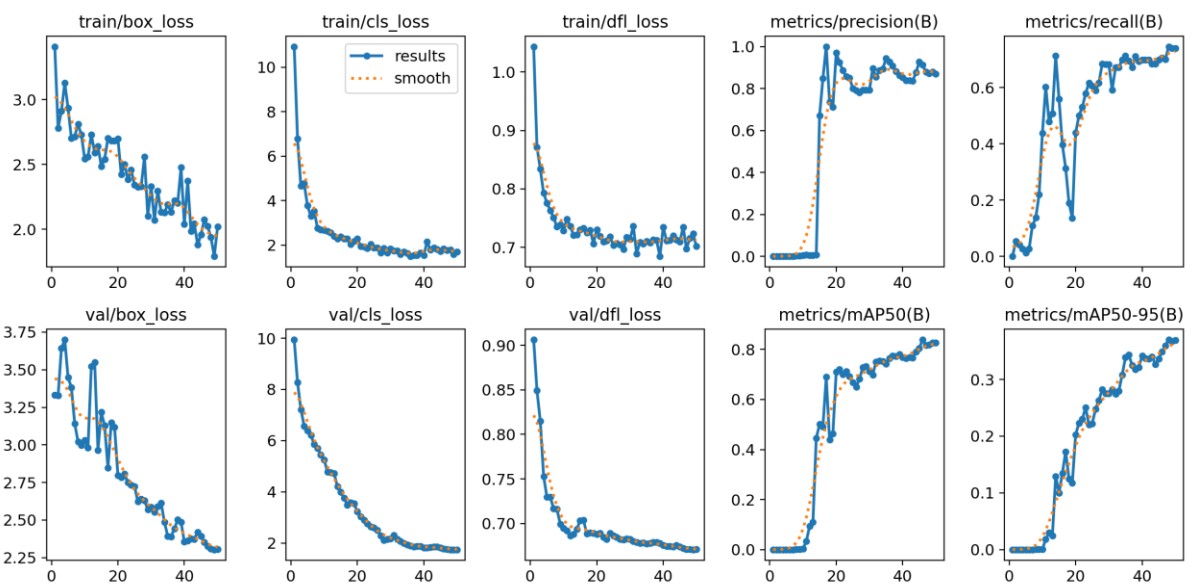
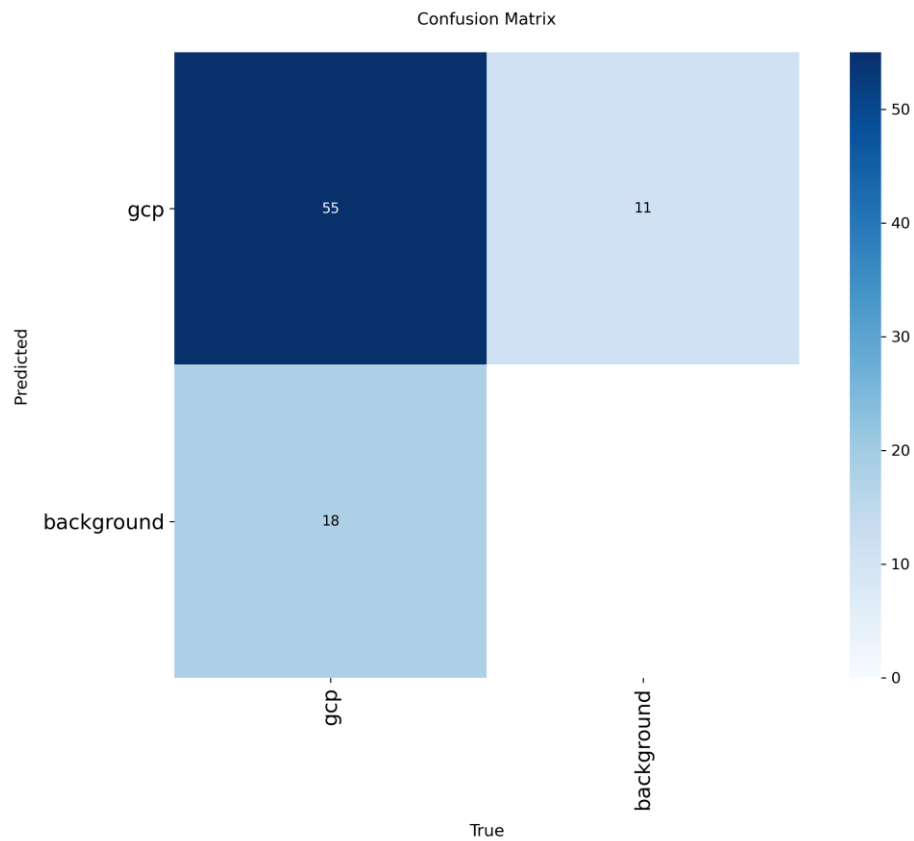
To fix this, we changed our approach and started using **tiled data**. This means we chopped the large images into smaller, overlapping pieces to keep the details sharp. When we tested our models on these smaller image tiles, the results got much better. We used this tiling method to test several models, ending with an older YOLO model, YOLOv5x being the best performing.

We tested Yolov8 and Yolov5 based on our research of previous similar projects, which they have used these models in. **This research can be seen in [MRR drones main research](#) document inside of Portflow.**

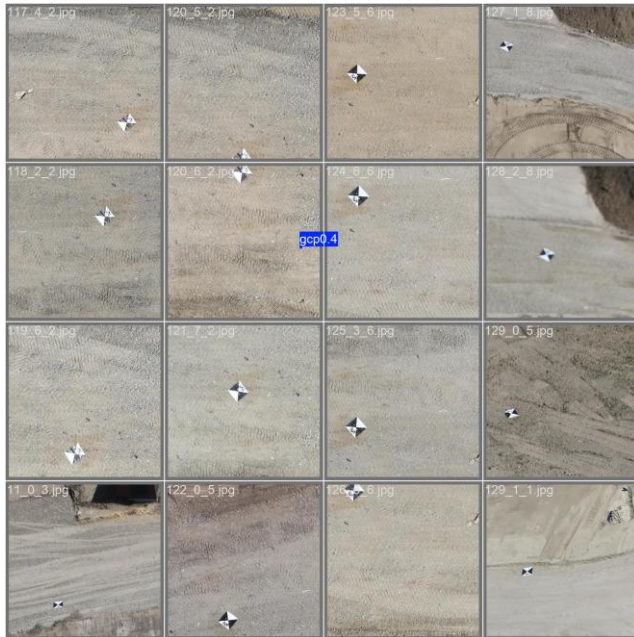
Results Matrix

Model Configuration	Dataset Strategy	Phase	Results / Performance Notes
YOLOv8n	Non-Tiled Data	Validation	Failed to make accurate predictions.
YOLOv8n	Non-Tiled Data	Testing	Failed completely. Discontinued this pipeline.
YOLOv8n	Tiled Data	Validation	Significant improvement over non-tiled baseline.
YOLOv8n	Tiled Data	Testing	Does not have good accuracy.
YOLOv5n	Tiled Data	Validation	Stable convergence; strong feature extraction on tiles.
YOLOv5n	Tiled Data	Testing	Best performing model from literature. Highly reliable predictions.
YOLOv5x	Tiled Data	Validation	
YOLOv5x	Tiled Data	Testing	Overall best performing model. Has the highest accuracy on predictions.
YOLO11x	Tiled Data	Testing	High rate of false positives. struggled to differentiate terrain features from actual targets.
YOLO26m (Latest)	Tiled Data	Validation	
YOLO26m (Latest)	Tiled Data	Testing	Not accurate. Performed the same as Yolov8n.

Validation yolov8n on non tiled data

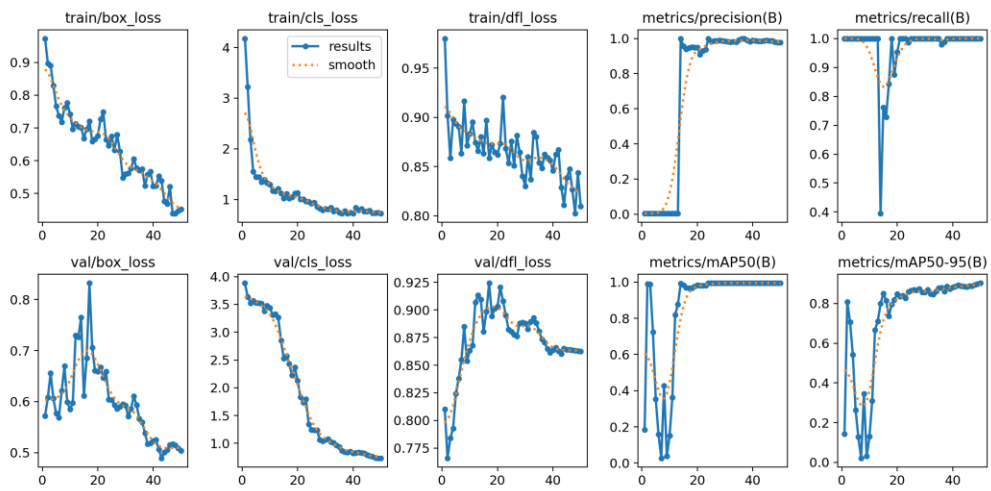
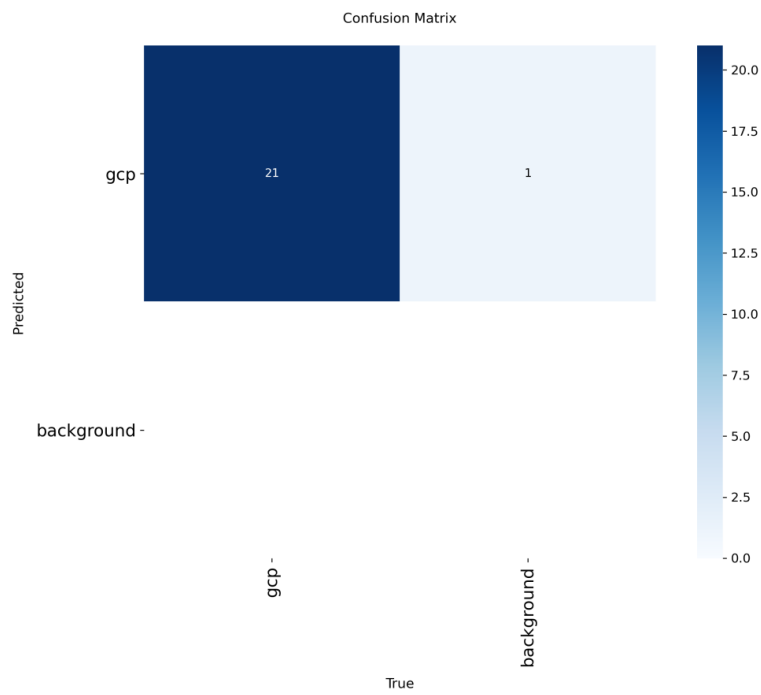


Testing yolov8n model on test set (non tiled data):



Yolov8n model does not perform well on non tiled data. So we decided to move along with tiled data on the following test.

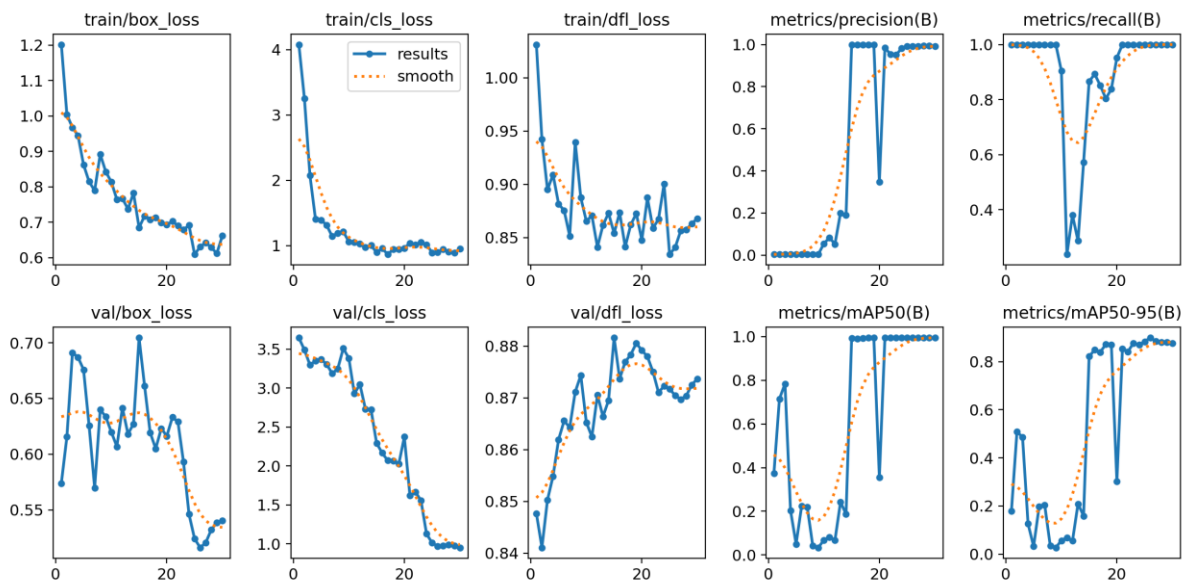
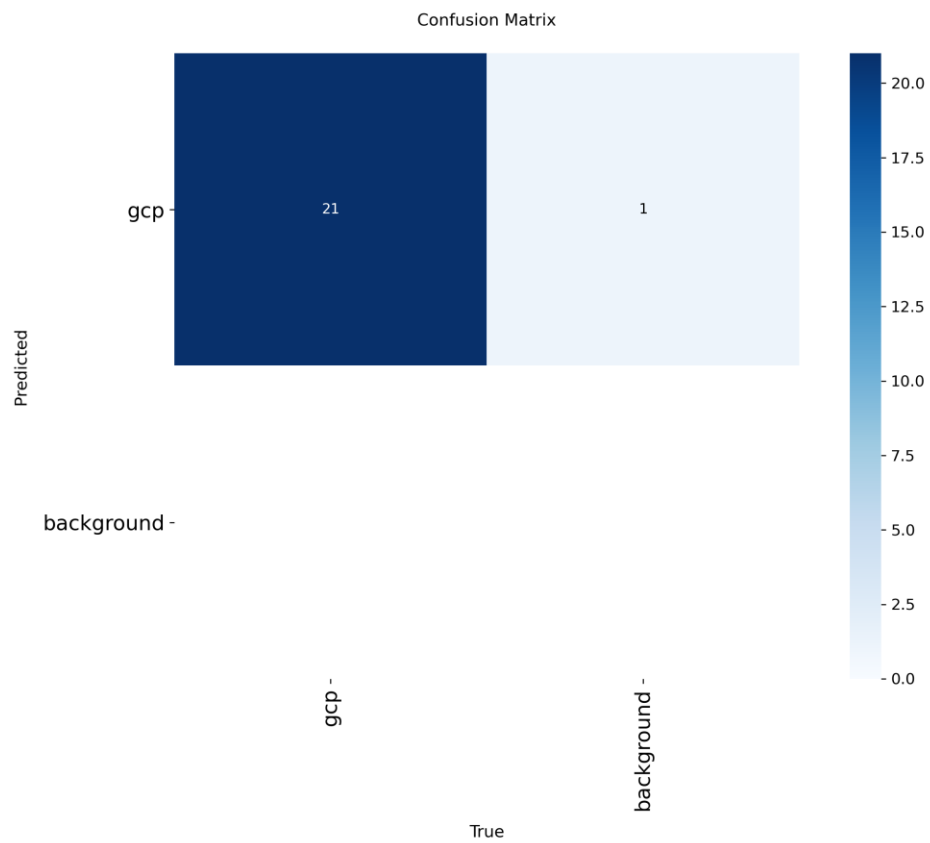
Validation yolov8n (tiled data):



Testing yolov8n model on test set(tiled data):



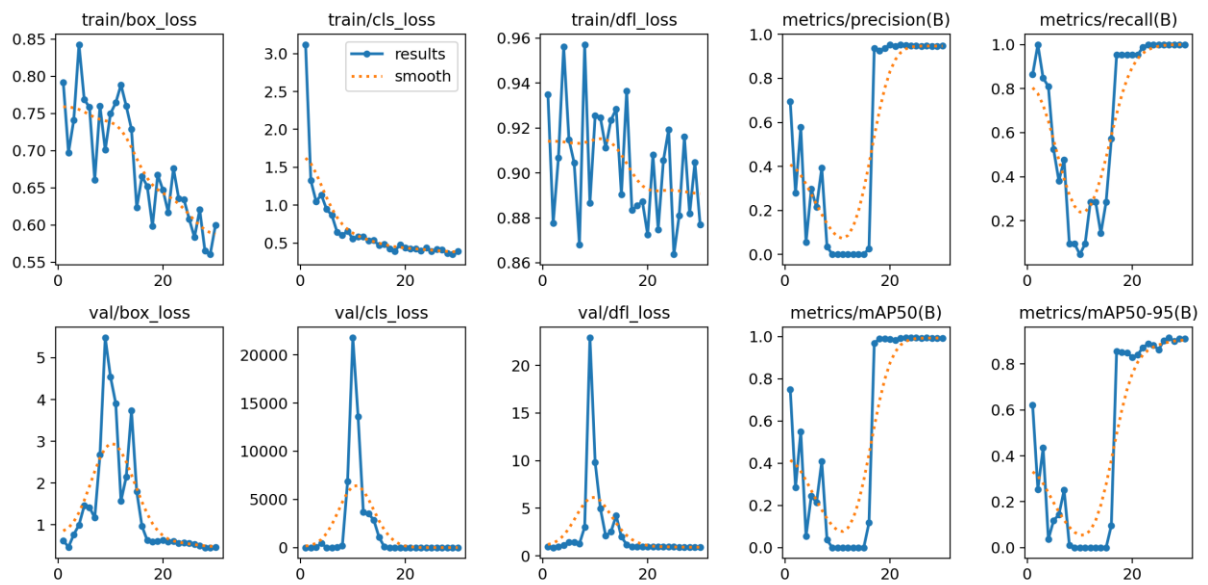
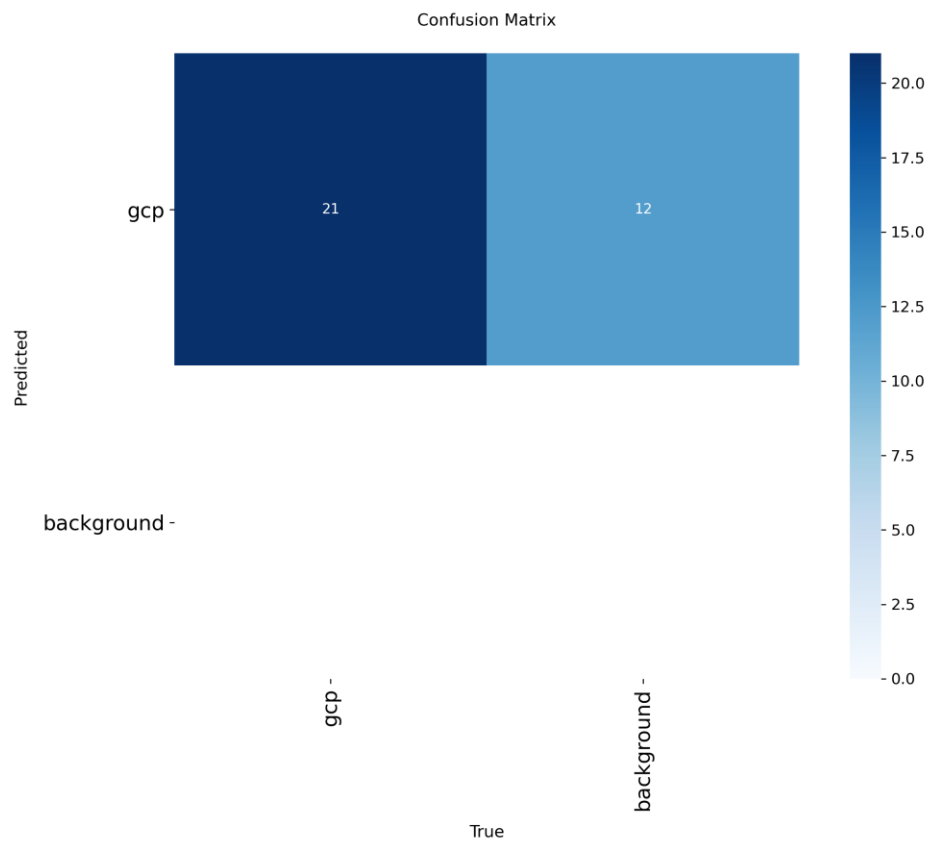
Validation yolov5n (tiled data):



Testing Yolov5n (tiled data):



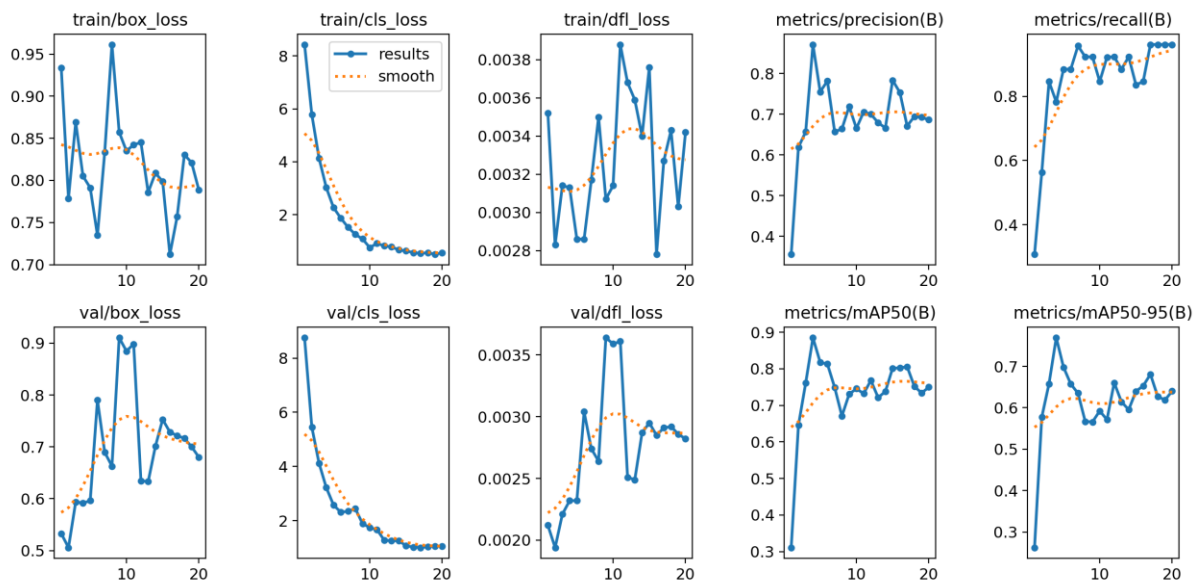
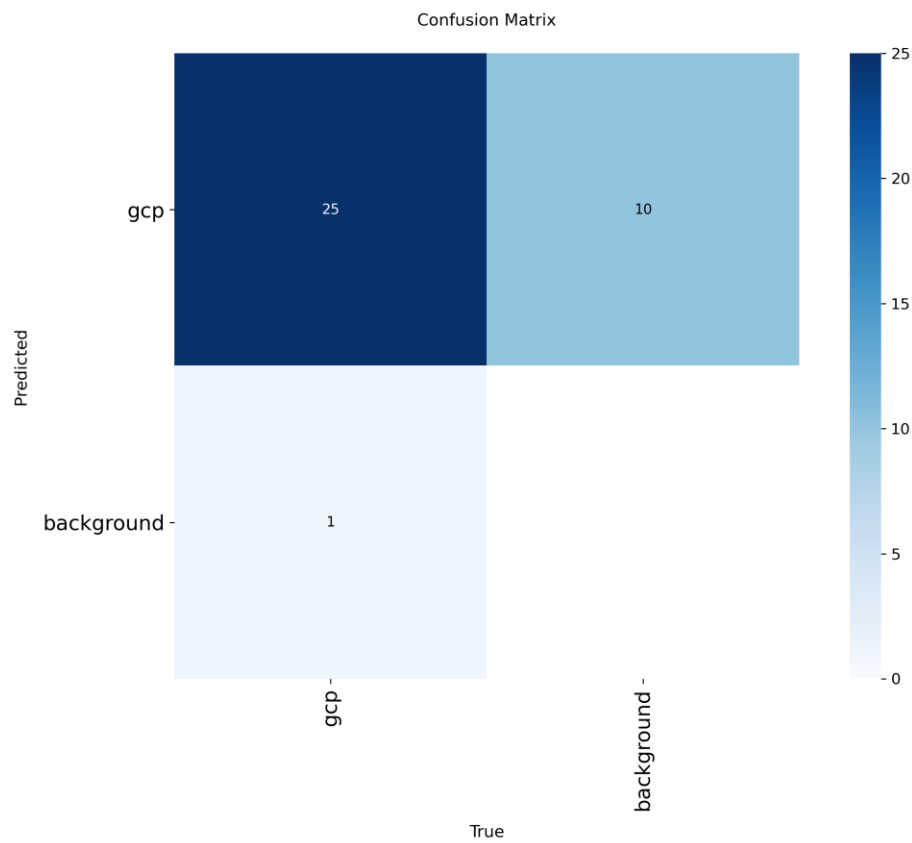
Validation yolov5x (tiled data)



Testing Yolov5x (tiled data):



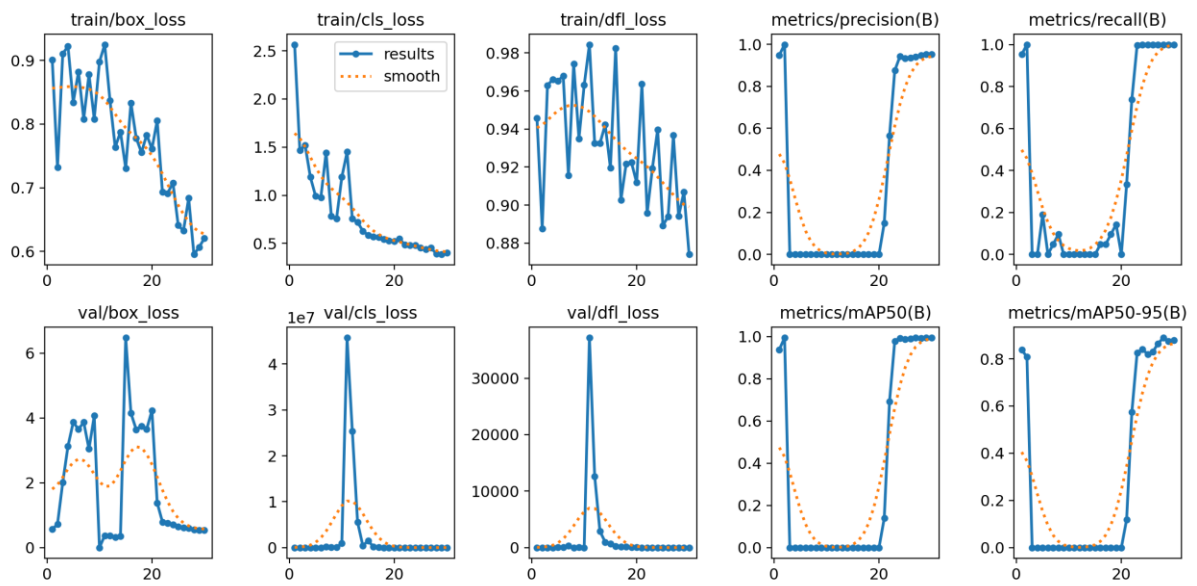
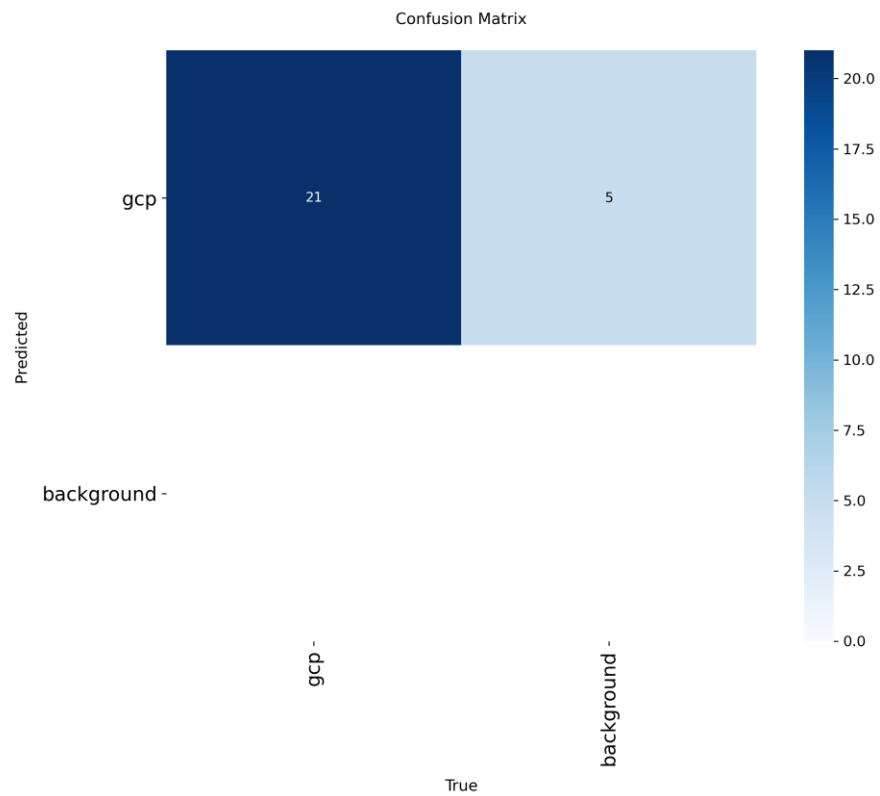
Validation Yolo26m model (Latest YOLO model)



Testing yolo26m model (tiled data)



Validation Yolo11x (tiled data)



Testing YOLO11X model (tiled data)

